The Eight-Fold Path to a Classical Pólya Urn Computation

Mathematical Foundations Series

Abstract

We consider the classical two-color Pólya urn started from 3 white balls and 2 black balls. At each step, one ball is drawn uniformly at random, returned to the urn, and an additional ball of the same color is added. If X_N denotes the proportion of white balls after N draws, we compute

$$P_\infty := \lim_{N\to\infty} \mathbb{P}\left(X_N > \frac{1}{2}\right).$$

The answer is

$$P_\infty = \frac{11}{16}.$$

This note presents eight rigorous proofs/viewpoints on the computation: an exact beta-binomial formula with explicit finite summation, a Riemann-sum limit, a Bayesian/conjugacy argument, a martingale-plus-factorial-moments proof, an exchangeability/de Finetti proof, a continuous-time branching-process embedding, a first-tie/Catalan decomposition, and an analytic generating-function approach based on a transport PDE together with a Beta-kernel normalization.

Contents

1 Introduction and notation **2**

2 Solution 1: Exact state probabilities and explicit finite summation **3**

3 Solution 2: A Riemann-sum asymptotic **5**

4 Solution 3: Bayesian conjugacy and predictive equivalence **6**

5 Solution 4: Martingale convergence and factorial moments **7**

6 Solution 5: Exchangeability and de Finetti's theorem **9**

7 Solution 6: Continuous-time embedding via Yule processes **10**

8 Solution 7: First tie, Catalan numbers, and a finite-time tie lemma **11**

9 Solution 8: Transport PDEs, raw weights, and Beta normalization **14**

10 Conclusion **16**

1 Introduction and notation

Let $W_0 = 3$ and $B_0 = 2$. After N draws, let

$$W_N = \text{number of white balls}, \qquad B_N = \text{number of black balls},$$

so that

$$W_N + B_N = N + 5.$$

Let

$$I_N := W_N - 3$$

be the number of white draws among the first N draws. Then

$$W_N = 3 + I_N, \qquad B_N = 2 + (N - I_N).$$

The white proportion is

$$X_N := \frac{W_N}{N+5} = \frac{3 + I_N}{N+5}.$$

We seek

$$P_\infty := \lim_{N\to\infty} \mathbb{P}\left(X_N > \frac{1}{2}\right).$$

Note that

$$X_N > \frac{1}{2} \iff \frac{3 + I_N}{N+5} > \frac{1}{2} \iff I_N > \frac{N-1}{2}.$$

For $a > 0$ and $m \in \mathbb{N}$, write

$$(a)_m := a(a+1)\cdots(a+m-1),$$

with $(a)_0 = 1$, for the rising factorial.

The eight routes below fall into three broad families:

- exact discrete formulas (Solutions 1, 7, 8),
- probabilistic representation theorems and stochastic-process arguments (Solutions 3, 4, 5, 6),
- asymptotic extraction (Solution 2).

Several of them identify the full limit law

$$X_\infty \sim \mathrm{Beta}(3, 2),$$

while Solution 1 derives the answer directly from the exact finite-N law. In the Beta-law routes, the desired probability is the Beta tail

$$\int_{1/2}^{1} 12x^2(1-x)\,dx = \frac{11}{16}.$$

Theorem 1. *For the Pólya urn started from* $(3, 2)$,

$$\lim_{N\to\infty} \mathbb{P}\left(X_N > \frac{1}{2}\right) = \frac{11}{16}.$$

2 Solution 1: Exact state probabilities and explicit finite summation

This gives the exact finite-N distribution and then evaluates the majority probability explicitly.

Step 1: probability of a specific color word

Fix a particular color sequence of length N containing exactly k white draws and $N - k$ black draws.

Along that path, the white-draw factors contributed to the numerator are

$$3 \cdot 4 \cdots (k + 2) = (3)_k,$$

and the black-draw factors are

$$2 \cdot 3 \cdots (N - k + 1) = (2)_{N-k}.$$

The denominator is always

$$5 \cdot 6 \cdots (N + 4) = (5)_N.$$

Hence every such color word has probability

$$\frac{(3)_k(2)_{N-k}}{(5)_N}.$$

Step 2: sum over all words with k white draws

There are $\binom{N}{k}$ such color words. Therefore

$$\mathbb{P}(I_N = k) = \binom{N}{k} \frac{(3)_k(2)_{N-k}}{(5)_N}.$$

This is the beta-binomial distribution with parameters $(N, 3, 2)$.

Expanding the factorials gives the concrete formula

$$\begin{aligned}\mathbb{P}(I_N = k) &= \binom{N}{k} \frac{(k + 2)!}{2!} \frac{(N - k + 1)!}{1!} \frac{4!}{(N + 4)!} \\ &= \frac{12\,(k + 1)(k + 2)(N - k + 1)}{(N + 1)(N + 2)(N + 3)(N + 4)}.\end{aligned}$$

Step 3: write the finite-N majority probability as a polynomial sum

Since

$$X_N > \frac{1}{2} \quad \Longleftrightarrow \quad I_N > \frac{N-1}{2},$$

we have

$$\mathbb{P}\left(X_N > \frac{1}{2}\right) = \sum_{k > \frac{N-1}{2}} \frac{12\,(k+1)(k+2)(N-k+1)}{(N+1)(N+2)(N+3)(N+4)}.$$

The denominator is constant in k, so the problem reduces to summing the cubic polynomial

$$(k+1)(k+2)(N-k+1).$$

Expanding,

$$(k+1)(k+2)(N-k+1) = -k^3 + (N-2)k^2 + (3N+1)k + (2N+2).$$

Step 4: evaluate the sum in the even case

Let $N = 2m$. Then

$$X_{2m} > \frac{1}{2} \quad \Longleftrightarrow \quad I_{2m} \geq m,$$

so

$$\mathbb{P}\left(X_{2m} > \frac{1}{2}\right) = \frac{12}{(2m+1)(2m+2)(2m+3)(2m+4)} \sum_{k=m}^{2m} (k+1)(k+2)(2m-k+1).$$

Using the standard formulas for $\sum k$, $\sum k^2$, and $\sum k^3$, one finds

$$\sum_{k=m}^{2m} (k+1)(k+2)(2m-k+1) = \frac{11m^4 + 58m^3 + 109m^2 + 86m + 24}{12}.$$

Therefore

$$\mathbb{P}\left(X_{2m} > \frac{1}{2}\right) = \frac{11m^2 + 25m + 12}{4(2m+1)(2m+3)}.$$

Step 5: evaluate the sum in the odd case

Let $N = 2m+1$. Then

$$X_{2m+1} > \frac{1}{2} \quad \Longleftrightarrow \quad I_{2m+1} \geq m+1,$$

so

$$\mathbb{P}\left(X_{2m+1} > \frac{1}{2}\right) = \frac{12}{(2m+2)(2m+3)(2m+4)(2m+5)} \sum_{k=m+1}^{2m+1} (k+1)(k+2)(2m+2-k).$$

Again using the standard power-sum formulas, one obtains

$$\sum_{k=m+1}^{2m+1}(k+1)(k+2)(2m+2-k)=\frac{11m^4+63m^3+133m^2+121m+36}{12}.$$

Hence

$$\mathbb{P}\left(X_{2m+1}>\frac{1}{2}\right)=\frac{11m^2+41m+36}{4(2m+3)(2m+5)}.$$

Step 6: take the limit

In the even case,

$$\mathbb{P}\left(X_{2m}>\frac{1}{2}\right)=\frac{11m^2+25m+12}{4(2m+1)(2m+3)}\longrightarrow\frac{11}{16}.$$

In the odd case,

$$\mathbb{P}\left(X_{2m+1}>\frac{1}{2}\right)=\frac{11m^2+41m+36}{4(2m+3)(2m+5)}\longrightarrow\frac{11}{16}.$$

Therefore

$$P_\infty=\lim_{N\to\infty}\mathbb{P}\left(X_N>\frac{1}{2}\right)=\frac{11}{16}.$$

3 Solution 2: A Riemann-sum asymptotic

This route derives the limit directly from the exact finite law.

Step 1: recover the exact mass function locally

As in Solution 1, any color word with exactly k white draws and $N-k$ black draws has probability

$$\frac{(3)_k(2)_{N-k}}{(5)_N},$$

and there are $\binom{N}{k}$ such words. Hence

$$\mathbb{P}(I_N=k)=\binom{N}{k}\frac{(3)_k(2)_{N-k}}{(5)_N}=\frac{12\,(k+1)(k+2)(N-k+1)}{(N+1)(N+2)(N+3)(N+4)}.$$

Step 2: pass to a density

Write $x=k/N$. Uniformly in $0\le k\le N$,

$$\mathbb{P}(I_N=k)=\frac{12}{N}x^2(1-x)+O\left(\frac{1}{N^2}\right).$$

Therefore

$$\mathbb{P}\left(X_N>\frac{1}{2}\right)=\sum_{k>\frac{N-1}{2}}\left[\frac{1}{N}\,12\left(\frac{k}{N}\right)^2\left(1-\frac{k}{N}\right)+O\left(\frac{1}{N^2}\right)\right].$$

Since there are $O(N)$ summands, the total error is $O(1/N)$. Thus

$$\mathbb{P}\left(X_N > \frac{1}{2}\right) \longrightarrow \int_{1/2}^{1} 12x^2(1-x)\,dx.$$

Step 3: evaluate the integral

We compute

$$12\int_{1/2}^{1}(x^2 - x^3)\,dx = 12\left[\frac{x^3}{3} - \frac{x^4}{4}\right]_{1/2}^{1} = \left[4x^3 - 3x^4\right]_{1/2}^{1} = 1 - \frac{5}{16} = \frac{11}{16}.$$

Hence

$$P_\infty = \frac{11}{16}.$$

4 Solution 3: Bayesian conjugacy and predictive equivalence

This route interprets the urn as a mixture of Bernoulli sequences.

Step 1: the latent parameter

Let $P \sim \mathrm{Beta}(3,2)$, with density

$$f(p) = \frac{p^{3-1}(1-p)^{2-1}}{B(3,2)} = 12p^2(1-p), \qquad 0 < p < 1.$$

Conditional on $P = p$, let the colors be i.i.d. Bernoulli(p), where "white" means success.

Step 2: posterior predictive probabilities

After observing k white draws in N trials, the posterior is

$$P \mid \text{data} \sim \mathrm{Beta}(3+k,\, 2+N-k).$$

Its mean is

$$\mathbb{E}[P \mid \text{data}] = \frac{3+k}{5+N}.$$

But this is exactly the physical probability of drawing a white ball from the urn when the current state is

$$(W_N, B_N) = (3+k,\, 2+N-k).$$

Therefore the Bayesian predictive process and the Pólya urn have the same finite-dimensional distributions.

Step 3: pass to the limit

Conditional on P, the strong law of large numbers gives

$$\frac{I_N}{N} \longrightarrow P \qquad \text{almost surely.}$$

Since

$$X_N = \frac{3 + I_N}{N+5} \quad \text{and} \quad \frac{I_N}{N} \to P,$$

we also have

$$X_N \longrightarrow P \qquad \text{almost surely.}$$

Thus

$$X_\infty \sim \mathrm{Beta}(3,2),$$

and

$$P_\infty = \mathbb{P}\left(P > \frac{1}{2}\right) = \int_{1/2}^{1} 12p^2(1-p)\,dp = \frac{11}{16}.$$

5 Solution 4: Martingale convergence and factorial moments

This proof identifies the limit law through martingales alone.

Step 1: X_N is a bounded martingale

Let $\mathcal{F}_N$ be the natural filtration. Since

$$\mathbb{P}(W_{N+1} = W_N + 1 \mid \mathcal{F}_N) = \frac{W_N}{N+5},$$

we have

$$\begin{aligned}\mathbb{E}[X_{N+1} \mid \mathcal{F}_N] &= \frac{W_N+1}{N+6} \cdot \frac{W_N}{N+5} + \frac{W_N}{N+6} \cdot \left(1 - \frac{W_N}{N+5}\right) \\ &= \frac{W_N}{N+5} = X_N.\end{aligned}$$

Thus (X_N) is a martingale. Since $0 \le X_N \le 1$, the martingale convergence theorem gives

$$X_N \longrightarrow X_\infty \qquad \text{almost surely and in } L^1.$$

Step 2: factorial-moment martingales

Fix $m \ge 1$ and define

$$M_N^{(m)} := \frac{(W_N)_m}{(N+5)_m}.$$

We claim $(M_N^{(m)})$ is also a martingale.

Indeed, given $\mathcal{F}_N$, either $W_{N+1} = W_N + 1$ with probability $W_N/(N+5)$, or $W_{N+1} = W_N$ with the complementary probability. Hence

$$\mathbb{E}[(W_{N+1})_m \mid \mathcal{F}_N] = \frac{W_N}{N+5}(W_N+1)_m + \left(1 - \frac{W_N}{N+5}\right)(W_N)_m.$$

Using

$$(W_N+1)_m = \frac{W_N+m}{W_N}(W_N)_m,$$

we obtain

$$\begin{aligned}\mathbb{E}[(W_{N+1})_m \mid \mathcal{F}_N] &= \left(\frac{W_N+m}{N+5} + 1 - \frac{W_N}{N+5}\right)(W_N)_m \\ &= \frac{N+5+m}{N+5}(W_N)_m.\end{aligned}$$

Therefore

$$\mathbb{E}[M_{N+1}^{(m)} \mid \mathcal{F}_N] = \frac{1}{(N+6)_m} \cdot \frac{N+5+m}{N+5}(W_N)_m = \frac{(W_N)_m}{(N+5)_m} = M_N^{(m)}.$$

Step 3: identify the moments of X_∞

Since $0 \le M_N^{(m)} \le 1$ and

$$\frac{(W_N)_m}{(N+5)_m} \longrightarrow X_\infty^m \qquad \text{almost surely,}$$

dominated convergence gives

$$\mathbb{E}[X_\infty^m] = \lim_{N\to\infty} \mathbb{E}[M_N^{(m)}] = \mathbb{E}[M_0^{(m)}] = \frac{(3)_m}{(5)_m}.$$

But these are exactly the moments of $\mathrm{Beta}(3,2)$:

$$\mathbb{E}[P^m] = \frac{(3)_m}{(5)_m}, \qquad P \sim \mathrm{Beta}(3,2).$$

Since probability measures on $[0,1]$ are uniquely determined by their moments (Hausdorff moment problem), it follows that

$$X_\infty \sim \mathrm{Beta}(3,2).$$

Therefore

$$P_\infty = \mathbb{P}\left(X_\infty > \frac{1}{2}\right) = \frac{11}{16}.$$

Remark 1. *There is also a different structural identification of the same Beta law via exchangeability and de Finetti's theorem. We present that separately in Solution 5.*

6 Solution 5: Exchangeability and de Finetti's theorem

This proof identifies the directing measure of the exchangeable color sequence.

Step 1: exchangeability from local path weights

Fix a color word of length N with exactly k white draws and $N-k$ black draws. As in the exact path calculation,

$$\mathbb{P}(\text{that word}) = \frac{(3)_k(2)_{N-k}}{(5)_N}.$$

This depends only on the counts $(k, N-k)$, not on the order of the colors. Therefore the infinite color sequence is exchangeable.

Step 2: de Finetti representation

By de Finetti's theorem, there exists a unique probability measure μ on $[0,1]$ such that for all N and k,

$$\mathbb{P}(I_N = k) = \binom{N}{k} \int_0^1 p^k (1-p)^{N-k}\, d\mu(p).$$

On the other hand, summing over the $\binom{N}{k}$ exchangeable words gives

$$\mathbb{P}(I_N = k) = \binom{N}{k} \frac{(3)_k(2)_{N-k}}{(5)_N}.$$

Hence

$$\int_0^1 p^k (1-p)^{N-k}\, d\mu(p) = \frac{(3)_k(2)_{N-k}}{(5)_N}.$$

Step 3: identify the directing measure

Now

$$\begin{aligned} 12 \int_0^1 p^{k+2}(1-p)^{N-k+1}\, dp &= 12\, B(k+3, N-k+2) \\ &= 12\, \frac{\Gamma(k+3)\Gamma(N-k+2)}{\Gamma(N+5)} \\ &= \frac{(3)_k(2)_{N-k}}{(5)_N}. \end{aligned}$$

Therefore μ is exactly the Beta$(3,2)$ law:

$$d\mu(p) = 12p^2(1-p)\, dp.$$

Conditional on the directing variable P, the colors are i.i.d. Bernoulli(P), so the empirical fraction of white draws converges almost surely to P. Therefore $X_N \to P$ almost surely, and hence

$$X_\infty \sim \text{Beta}(3,2).$$

Thus

$$P_\infty = \frac{11}{16}.$$

7 Solution 6: Continuous-time embedding via Yule processes

This route embeds the urn into continuous time.

Step 1: the continuous-time model

Attach an independent rate-1 exponential clock to each ball. When a ball's clock rings, that ball produces one offspring of the same color. Let $W(t)$ and $B(t)$ be the white and black populations at time t.

At any moment, the total birth rate is $W(t) + B(t)$, and the probability that the next birth is white is

$$\frac{W(t)}{W(t) + B(t)}.$$

Therefore, if one observes the process only at birth times, the embedded jump chain is exactly the original Pólya urn.

Step 2: Yule-process limits

The white and black populations evolve independently as Yule processes started from 3 and 2 individuals. A standard theorem for Yule processes states that

$$e^{-t}W(t) \longrightarrow \mathcal{W}, \qquad e^{-t}B(t) \longrightarrow \mathcal{B} \qquad \text{almost surely,}$$

where

$$\mathcal{W} \sim \text{Gamma}(3, 1), \qquad \mathcal{B} \sim \text{Gamma}(2, 1),$$

and $\mathcal{W}$ and $\mathcal{B}$ are independent.

Step 3: pass to the ratio

Let τ_N be the time of the N-th birth. Then $\tau_N \to \infty$ almost surely and

$$W(\tau_N) = W_N, \qquad B(\tau_N) = B_N.$$

Hence

$$X_N = \frac{W_N}{W_N + B_N} = \frac{W(\tau_N)}{W(\tau_N) + B(\tau_N)} \longrightarrow \frac{\mathcal{W}}{\mathcal{W} + \mathcal{B}} \qquad \text{almost surely.}$$

By the gamma-beta algebra, if $U \sim \Gamma(\alpha, 1)$ and $V \sim \Gamma(\beta, 1)$ are independent, then

$$\frac{U}{U + V} \sim \text{Beta}(\alpha, \beta).$$

Therefore

$$X_\infty \sim \mathrm{Beta}(3,2),$$

and

$$P_\infty = \frac{11}{16}.$$

8 Solution 7: First tie, Catalan numbers, and a finite-time tie lemma

This proof is self-contained and works directly with the finite-time probabilities

$$\mathbb{P}\left(X_N > \frac{1}{2}\right),$$

without needing any continuity statement about the limiting law.

Step 1: majority probabilities from a tied start

For $m \geq 1$, let $q_N^{(m)}$ denote the probability that, starting from a tied state (m,m), the white proportion is strictly larger than $1/2$ after N further draws.

Lemma 1 (tie lemma)**.** *For each fixed $m \geq 1$,*

$$q_N^{(m)} \longrightarrow \frac{1}{2} \qquad \text{as } N \to \infty.$$

More precisely:

- *if $N = 2n+1$ is odd, then $q_{2n+1}^{(m)} = \frac{1}{2}$ exactly;*
- *if $N = 2n$ is even, then*

$$q_{2n}^{(m)} = \frac{1}{2}\left(1 - \binom{2n}{n}\frac{(m)_n^2}{(2m)_{2n}}\right),$$

  and the tie probability

$$\binom{2n}{n}\frac{(m)_n^2}{(2m)_{2n}} \longrightarrow 0.$$

Proof. Start from (m,m).

If $N = 2n+1$ is odd, then after $2n+1$ draws the difference between white and black counts is odd, so a tie is impossible. By color symmetry,

$$q_{2n+1}^{(m)} = \frac{1}{2}.$$

If $N = 2n$ is even, then white has strict majority exactly when there are more white draws than black draws among these $2n$ draws. By symmetry,

$$q_{2n}^{(m)} = \frac{1}{2}\Big(1 - \mathbb{P}(\text{tie after } 2n \text{ draws from } (m,m))\Big).$$

A tie after $2n$ draws means exactly n white draws and n black draws. Every such color word has probability

$$\frac{(m)_n(m)_n}{(2m)_{2n}},$$

and there are $\binom{2n}{n}$ such words. Therefore

$$\mathbb{P}(\text{tie after } 2n \text{ draws from } (m,m)) = \binom{2n}{n}\frac{(m)_n^2}{(2m)_{2n}}.$$

Using Stirling's formula,

$$\binom{2n}{n}\frac{(m)_n^2}{(2m)_{2n}} = \binom{2n}{n}\frac{\Gamma(m+n)^2\Gamma(2m)}{\Gamma(m)^2\Gamma(2m+2n)} = O(n^{-1/2}),$$

so it tends to 0. Hence

$$q_{2n}^{(m)} \longrightarrow \frac{1}{2}.$$

Combining the odd and even cases proves the lemma. □

Step 2: the first-tie distribution from the initial state $(3,2)$

Now return to the original urn started from $(3,2)$, and let

$$T := \inf\{N \geq 0 : W_N = B_N\}$$

be the first time the process ever reaches a tie.

A first tie can only occur at some state

$$(n+3, n+3), \qquad n \geq 0,$$

after exactly $2n+1$ draws, consisting of

$$n \text{ white draws and } n+1 \text{ black draws.}$$

To be a *first* tie, the path must stay strictly above the diagonal until the final step. The number of such paths is the Catalan number

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

Every such path has probability

$$\frac{(3)_n(2)_{n+1}}{(5)_{2n+1}} = 12\,\frac{(n+2)!^2}{(2n+5)!}.$$

Therefore

$$a_n := \mathbb{P}\big(T = 2n+1,\ (W_T, B_T) = (n+3, n+3)\big) = 12\,C_n\,\frac{(n+2)!^2}{(2n+5)!}.$$

Using

$$C_n = \frac{(2n)!}{n!(n+1)!},$$

this simplifies to

$$a_n = \frac{3(n+2)}{(2n+1)(2n+3)(2n+5)}.$$

Step 3: sum the first-tie probabilities

A partial fraction decomposition gives

$$\frac{3(n+2)}{(2n+1)(2n+3)(2n+5)} = \frac{9}{16(2n+1)} - \frac{3}{8(2n+3)} - \frac{3}{16(2n+5)}.$$

Hence

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{3(n+2)}{(2n+1)(2n+3)(2n+5)} = \frac{5}{8}.$$

So

$$\mathbb{P}(T < \infty) = \frac{5}{8}.$$

Step 4: decompose the finite-N majority event

For each $N \geq 0$,

$$\mathbb{P}_{(3,2)}\left(X_N > \frac{1}{2}\right) = \mathbb{P}(T > N) + \sum_{\substack{n \geq 0 \\ 2n+1 \leq N}} a_n \, q_{N-(2n+1)}^{(n+3)}.$$

Indeed:

- if $T > N$, then no tie has yet occurred, so the process is still strictly above the diagonal at time N, hence $X_N > 1/2$;
- if the first tie occurs at time $2n+1$, then the process is at the tied state $(n+3, n+3)$, and the remaining probability of white majority after $N - (2n+1)$ further draws is exactly $q_{N-(2n+1)}^{(n+3)}$.

Now let $N \to \infty$. Since

$$\mathbb{P}(T > N) \longrightarrow \mathbb{P}(T = \infty) = 1 - \sum_{n=0}^{\infty} a_n,$$

and since each $q_{N-(2n+1)}^{(n+3)} \to \frac{1}{2}$ by the tie lemma, dominated convergence yields

$$\begin{aligned} P_\infty &= \lim_{N\to\infty} \mathbb{P}_{(3,2)}\left(X_N > \frac{1}{2}\right) \\ &= \mathbb{P}(T = \infty) + \sum_{n=0}^{\infty} a_n \cdot \frac{1}{2} \\ &= \left(1 - \sum_{n=0}^{\infty} a_n\right) + \frac{1}{2}\sum_{n=0}^{\infty} a_n \\ &= 1 - \frac{1}{2}\sum_{n=0}^{\infty} a_n. \end{aligned}$$

Since $\sum a_n = 5/8$, we obtain

$$P_\infty = 1 - \frac{1}{2} \cdot \frac{5}{8} = \frac{11}{16}.$$

9 Solution 8: Transport PDEs, raw weights, and Beta normalization

This analytic route begins with an unnormalized generating function whose coefficients capture the correct state weights up to a factor depending only on the total level $N = i + j$. A Beta-kernel integral then restores the exact normalization.

Step 1: the unnormalized generating function

For $i, j \geq 0$, define the raw weight

$$u_{i,j} := \frac{(3)_i (2)_j}{i!\, j!} = \binom{i+2}{2}(j+1).$$

Consider the shifted generating function

$$U(x, y, t) := \sum_{i,j \geq 0} u_{i,j}\, x^{i+3} y^{j+2} t^{i+j}.$$

Summing the geometric series gives

$$U(x, y, t) = x^3 y^2 \sum_{i \geq 0} \binom{i+2}{2} (xt)^i \sum_{j \geq 0} (j+1)(yt)^j = \frac{x^3}{(1-xt)^3} \frac{y^2}{(1-yt)^2}.$$

Step 2: the transport PDE

The function U satisfies the first-order PDE

$$\frac{\partial U}{\partial t} = x^2 \frac{\partial U}{\partial x} + y^2 \frac{\partial U}{\partial y}, \qquad U(x, y, 0) = x^3 y^2.$$

Indeed, this follows either by direct differentiation of the closed form above or by the method of characteristics: along the characteristic curves

$$\frac{dx}{dt} = -x^2, \qquad \frac{dy}{dt} = -y^2,$$

the quantity U remains constant, yielding exactly

$$U(x, y, t) = \frac{x^3}{(1-xt)^3} \frac{y^2}{(1-yt)^2}.$$

The coefficient of $x^{i+3} y^{j+2} t^{i+j}$ in U is $u_{i,j}$, which captures the correct state dependence in i and j but is not yet normalized to a probability.

Step 3: identify the missing normalization

The exact probability from Solution 1 is

$$p_{i,j} := \mathbb{P}(I_{i+j} = i) = \binom{i+j}{i} \frac{(3)_i (2)_j}{(5)_{i+j}}.$$

Since

$$\binom{i+j}{i} \frac{(3)_i (2)_j}{(5)_{i+j}} = u_{i,j} \cdot \frac{(i+j)!}{(5)_{i+j}},$$

the discrepancy between $u_{i,j}$ and $p_{i,j}$ depends only on the total level

$$N = i + j.$$

For this urn,

$$\frac{N!}{(5)_N} = \frac{24}{(N+1)(N+2)(N+3)(N+4)} = 4 \int_0^1 u^N (1-u)^3 \, du.$$

This suggests the integral operator

$$(\mathcal{T} f)(t) := 4 \int_0^1 (1-u)^3 f(ut) \, du,$$

which multiplies the coefficient of t^N by the factor $N!/(5)_N$.

Step 4: Beta-kernel normalization

Define the normalized shifted generating function

$$\widetilde{G}(x, y, t) := (\mathcal{T} U)(x, y, t) = 4 \int_0^1 (1-u)^3 U(x, y, ut) \, du.$$

Substituting the closed form for U gives

$$\widetilde{G}(x, y, t) = 4x^3 y^2 \int_0^1 \frac{(1-u)^3}{(1-uxt)^3 (1-uyt)^2} \, du.$$

Now extract coefficients under the integral:

$$\begin{aligned}
[x^{i+3} y^{j+2} t^{i+j}] \, \widetilde{G} &= 4 \binom{i+2}{2} (j+1) \int_0^1 u^{i+j} (1-u)^3 \, du \\
&= 4 \binom{i+2}{2} (j+1) B(i+j+1, 4) \\
&= \frac{12(i+1)(i+2)(j+1)}{(i+j+1)(i+j+2)(i+j+3)(i+j+4)} \\
&= p_{i,j}.
\end{aligned}$$

Thus

$$\widetilde{G}(x, y, t) = \sum_{i,j \geq 0} p_{i,j} \, x^{i+3} y^{j+2} t^{i+j}.$$

In particular, the PDE route reproduces the exact beta-binomial law.

Step 5: recover the Beta-mixture form

If one removes the harmless initial factor x^3y^2, the corresponding unshifted normalized generating function is

$$G(x,y,t) := \sum_{i,j\geq 0} p_{i,j}\, x^i y^j t^{i+j}.$$

Using the Beta-binomial formula and summing over $N = i + j$,

$$\begin{aligned} G(x,y,t) &= \sum_{N\geq 0}\sum_{i=0}^{N} \binom{N}{i} \left(\int_0^1 12\, p^{i+2}(1-p)^{N-i+1}\, dp\right) x^i y^{N-i} t^N \\ &= 12 \int_0^1 p^2(1-p) \sum_{N\geq 0} \big(t(px + (1-p)y)\big)^N\, dp \\ &= 12 \int_0^1 \frac{p^2(1-p)}{1 - t(px + (1-p)y)}\, dp. \end{aligned}$$

This is exactly the generating-function form of the Beta$(3,2)$ mixture.

Therefore the coefficient array is beta-binomial, and the same limit law follows:

$$X_\infty \sim \mathrm{Beta}(3,2).$$

Hence

$$P_\infty = \mathbb{P}\left(X_\infty > \frac{1}{2}\right) = \int_{1/2}^1 12p^2(1-p)\, dp = \frac{11}{16}.$$

Remark 2 (optional normalized PDE)**.** *One can also write down a first-order PDE directly for the normalized shifted generating function $\widetilde{G}$:*

$$t\,\widetilde{G}_t + 4\widetilde{G} - 4x^3y^2 = tx^2\widetilde{G}_x + ty^2\widetilde{G}_y, \qquad \widetilde{G}(x,y,0) = x^3y^2.$$

This equation is exact, but less transparent than the raw transport PDE for U. For this reason, the unnormalized PDE together with the Beta-kernel normalization is the cleaner analytic route.

10 Conclusion

All eight routes lead to the same answer:

$$\boxed{\lim_{N\to\infty} \mathbb{P}\left(X_N > \frac{1}{2}\right) = \frac{11}{16}}.$$

The common structural theme is the repeated appearance of the Beta$(3,2)$ law:

- exactly, via the beta-binomial formula;
- asymptotically, via the density $12x^2(1-x)$;
- probabilistically, via Bayesian conjugacy, martingales, de Finetti, and Yule processes;
- analytically, via a transport PDE for raw weights and a Beta-kernel normalization.